
Goodreads Data Analysis using Spark, Airflow, and Docker

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Outline

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Overview Project

Overview

Project Title : Goodreads Data Analysis using Spark, Airflow, and Docker.

Project Goal : To efficiently process Goodreads data to uncover meaningful insights and providing actionable recommendations to various stakeholders.

Analysis : Top-rated books per year, Most-reviewed books per year, Most-rated books per year. Correlation between Average Rating and Number of Pages and Correlation between Average Rating and Publication Year.

Outcomes:

- The project has successfully implemented Docker, Apache Airflow, Apache Spark to process the Goodreads data efficiently.

Based on the Analysis the outcomes are :

- Generate valuable insights into reading trends, including the most reviewed books, top rated books, most rated books.
- The project enhances the reading experience for readers, helping readers discover books aligned with readers preferences.
- The analysis offers actionable recommendations for authors and publishers, helping them understand readers preferences and market dynamics.

Technologies Used: Docker for containerized architecture, Apache Airflow for data workflow orchestration and Apache Spark for processing the Goodreads data.

Main Project

1. Project Background

- Project Title : Goodreads Data Analysis using Spark, Airflow, and Docker.
- The purpose of this project is to efficiently process Goodreads data to uncover meaningful insights and providing actionable recommendations to various stakeholders.

2. Problem Statement

This project addresses the challenge of efficiently processing Goodreads data to uncover meaningful insights and actionable recommendations.

To achieve this, It is necessary to implement a containerized architecture using Docker, Orchestration the data workflow with Apache Airflow, and utilize Apache Spark for efficient data (Goodreads Data) processing.

This project will focus on calculating specific metrics to derive insights from the data gives actionable recommendations to various stakeholders, including:

- Top-rated books per year.
- Most-reviewed books per year.
- Most-rated books per year.
- Correlation Analysis, specifically:
 - Correlation between Average Rating and Number of Pages.
 - Correlation between Average Rating and Publication Year.

3. Data Platform Understanding

- This project employs a robust containerized architecture using **Docker**, ensuring consistent deployment across environments and minimizing compatibility issues.
- The data workflow is orchestrated with **Apache Airflow**, which manages scheduling, monitoring, and execution of tasks in the processing pipeline.
- **Data Ingestion and Storage:** Data is sourced from **CSV** files containing various book attributes from Goodreads. After extraction, it is processed and cleaned with **Apache Spark**, then stored in a **PostgreSQL** database.
- **Orchestration with Airflow:** Apache Airflow, running in its own Docker container, schedules and manages the data pipeline, from extraction to storage. Its DAG structure clearly represents workflows and dependencies, enhancing maintainability and scalability.
- **Data Processing with Spark:** Apache Spark is the core engine for efficiently processing and cleaning data. Its distributed computing capabilities enable parallel processing of large datasets.

4. Data Understanding

Goodreads book dataset is sourced from

Kaggle:

<https://www.kaggle.com/datasets/jealousleopard/goodreadsbooks>

Size : 1.56 mb

Number of rows : 11127

Number of columns : 12

The dataset comprises various attributes related to books, sourced from Goodreads CSV files.

Key fields in the dataset include:

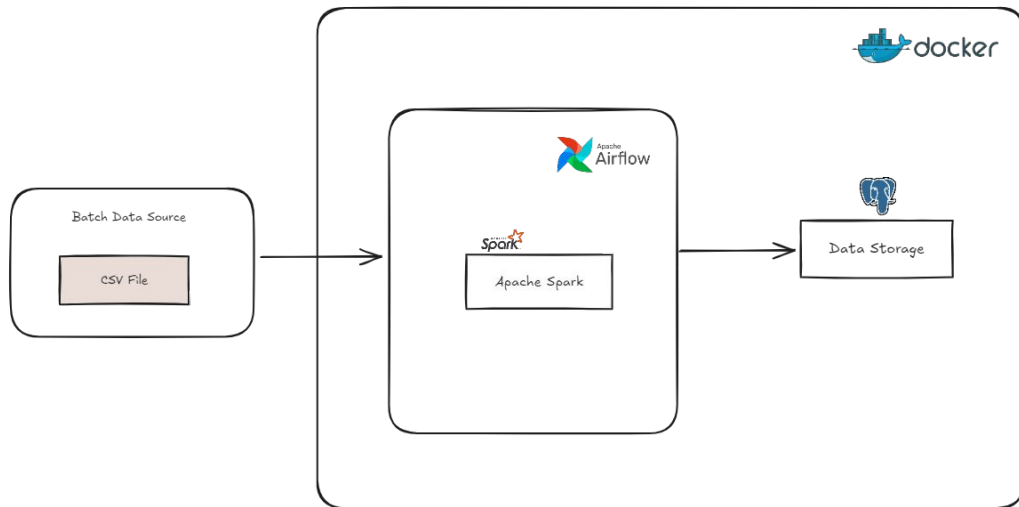
- BookID: Unique identifier for each book.
- Title: The title of the book.
- Authors: The authors of the book.
- Average Rating: The average rating given by users.
- ISBN : A 10-digit unique identifier for books.
- ISBN13 : A 13-digit updated version of ISBN for global use.
- Language Code : A standardized code indicating the book's language.
- Number of Pages: The total pages in the book.
- Ratings Count: The total number of ratings received.
- Text Reviews Count: The number of text reviews provided by users.
- Publication Date: The date the book was published.
- Publisher: The publisher of the book.

5. Project Architecture

The data processing pipeline was designed to handle batch processing.

The architecture includes :

- Docker for containerized architecture.
- Data Ingestion, loading the CSV file into Apache Spark.
- Processing the data using Apache Spark.
- Data Output, writing the transformed data to PostgreSQL for storage.
- Apache Airflow for data workflow orchestration.



5. Transformation

Apache Spark was selected as the core data processing tool because it excels in large-scale distributed data processing. **Transformations included:**

1. **Drop irrelevant columns for analysis** such as isbn and isbn13.
2. **Filtering invalid records**, such as those with non-numeric average_rating or missing ratings_count.
3. **Creating new features** such as publication_year by extracting it from the publication_date column.
4. **Standardizing string fields** (title, authors, and publisher) to ensure consistency and facilitate comparison.

The transformed data was **cached** then used for **various analyses**, such as **calculating correlations and ranking books**.

6. Data Modelling (Business)

The screenshot shows a database management interface. On the left, a 'Database Navigator' pane displays a schema named 'public' under a 'warehouse' database. The schema contains several tables: 'correlation_rating_num_pages' (8K), 'correlation_rating_publication_year' (8K), 'most_reviewed_books_per_year' (48K), and 'top_rated_books_per_year' (56K). Below this, a 'Project - General' pane shows 'Name' and 'DataSource'. The main window displays the 'top_rated_books_per_year' table with columns: 'bookID', 'title', 'authors', 'average_rating', and 'language_code'. The table contains 23 records of books, with the fourth record highlighted.

bookID	title	authors	average_rating	language_code
1	38,568 A QUICK BITE (ARGENEAU #1)	LYNSAY SANDS	3.91	ENG
2	1,337 ENTHUSIASM AND DIVINE MADNESS	JOSEF PIEPER/RICHARD WINSTON/CLARA WII	4.5	ENG
3	11,625 ARIEL: THE RESTORED EDITION	SYLVIA PLATH/FRIEDA HUGHES	4.27	ENG
4	21,266 RICK STEVES' EUROPE THROUGH THE BACK D	RICK STEVES	4.24	ENG
5	40,343 THE FULL BOX (FULL #1-4)	JANET EVANOVICH/CHARLOTTE HUGHES	4.29	ENG
6	41,179 MIRACLEMAN BOOK THREE: OLYMPUS	ALAN MOORE/JOHN TOTLEBEN/STEVE OLIFF/	4.39	ENG
7	30,117 WHERE THE SIDEWALK ENDS: POEMS AND DI	SHEL SILVERSTEIN	4.3	ENG
8	31,420 THE JOURNALS OF SYLVIA PLATH	SYLVIA PLATH/KAREN V. KUKIL	4.31	ENG
9	30 J.R.R. TOLKIEN 4-BOOK BOXED SET: THE HOBB	J.R.R. TOLKIEN	4.59	ENG
10	36,399 CHILDREN'S CHRISTMAS STORIES AND POEM	BOB BLAISDELL/PAT RONSON STEWART	4.5	ENG
11	12,240 THE PLAY SOLDIER	CHET GREEN	4.5	ENG
12	35,561 THE BEATLES' STORY ON CAPITOL RECORDS	F BRUCE SPIZER	4.72	ENG
13	2,545 THE TREASURED WRITINGS OF KAHLIL GIBRA	KAHLIL GIBRAN	4.42	ENG
14	20,954 WINCHESTER SHOTGUNS	DENNIS ADLER/R.L. WILSON	5	ENG
15	36,492 OLIVER WENDELL HOLMES IN PARIS: MEDIC	WILLIAM C. DOWLING	5	ENG
16	39,829 HIS PRINCESS DEVOTIONAL: A ROYAL ENCO	SHERI ROSE SHEPHERD	5	ENG
17	2,560 WILLEM DE KOONING: LATE PAINTINGS	JULIE SYLVESTER/DAVID SYLVESTER	5	ENG
18	17,224 THE DIAMOND COLOR MEDITATION: COLOR	JOHN DIAMOND	5	ENG
19	20,957 COLOSSIANS AND PHILEMON: A CRITICAL AN	R. MCL. WILSON	5	ENG
20	18,184 THE COMPLETE THEORY FUN FACTORY: MUSI	IAN MARTIN/KATIE ELLIOTT	5	ENG
21	32,823 FANNING THE FLAME: BIBLE CROSS AND MIS	CHRIS GREEN/CHRIS WRIGHT/PAUL DOUGLA	5	ENG
22	36,853 TYRANNOSAURUS WRECKS (STANLEY #1)	LAURA DRISCOLL/ALISA KLAYMAN-GRODSKY	5	ENG
23	38,804 THE IRISH ANATOMIST: A STUDY OF FLANN C	KEITH DONOHUE	5	ENG

The business logic for the project focused on creating tables that serve specific analytical purposes, such as:

1. Top-rated books per year
2. Most-reviewed books per year
3. Most-rated books per year
4. Correlation Analysis: A table summarizing the correlation coefficients between various numerical attributes in the dataset, specifically:
 - Correlation between Average Rating and Number of Pages.
 - Correlation between Average Rating and Publication Year.

7. Conclusion & Recommendation

Conclusion

- The project has successfully implemented Docker, Apache Airflow, Apache Spark to process the Goodreads data efficiently.

Based on the Analysis the outcomes are :

- Generate valuable insights into reading trends, including the most reviewed books, top rated books, most rated books.
- The project enhances the reading experience for readers, helping readers discover books aligned with readers preferences
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Recommendations

Scalability for real-time analytics, process real-time data from Goodreads.

Integrate with additional data sources, such as social media to provide deeper insights into user preferences and book trends.

Visualization tools could be used to make insights.

THANK YOU :)